

Technical Specifications for

CK Creek Pasture Fence

BUREAU OF LAND MANAGEMENT
Malta Field Office
501 South Second Street East
Malta, Montana 59538



March 2026

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CK CREEK PASTURE FENCE

PROJECT SPECIFICATIONS

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PART 1: GENERAL

1.01 SUMMARY:

- A. Description of Work: Furnishing labor, equipment, supplies, and materials to construct approximately 2.81 miles (900 rods) of new fence in accordance with these specifications, the Work Data Sheet, and the drawings. The new fence shall be 4-wire (3 barbed wires with bottom barbless wire), with all steel line posts (T-posts) except single set wood posts as noted in the drawings. Install cattle guards with wings at two locations as shown in drawings. Install sage grouse fence markers on full length of fence.
- B. Location: Work under this contract is in south Phillips County, State of Montana. The work location is shown on the maps and drawings.
- C. Work Data Sheet: Designates the type and quantity of work required at each project location, and dimensions required for structures as shown on typical drawings.

1.02 REFERENCES:

- A. Referenced Specifications/Standards with Abbreviations and/or Acronyms: Wherever the following acronyms are used in these specifications or in the drawings, they are to be construed the same as the respective expressions represented. Copies of the referenced specifications/standards referred to herein may be procured by the Contractor, from the following:

ASTM	American Society for Testing and Materials P.O. Box C700 West Conshohocken, PA 19248-2959
AWPA	American Wood Protection Association P.O. Box 361784 Birmingham, AL 35236-1784
AASHTO	American Association of State Highway and Transportation Officials 555 12 th Street NW Washington, DC 20004
AWS	American Welding Society 8669 NW 36 Street, #130 Miami, FL 33166-6672

1.03 DEFINITIONS:

- A. Approvals: Approval of the submittals is an indication that the Contractor's submittals have been reviewed and that there are no objections, except as noted. Approval of deviations shall apply only to those deviations or omissions from the requirements of the drawings and specifications brought to the Contracting Officer's attention in writing. After approval of an item, the Contractor may submit a

substitute for approval when the approved item cannot be purchased or delivered in time to avoid delay in completion of the project.

1.04 SUBMITTAL:

- A. Scope: Submittals shall be based on the requirements listed in these contract specifications. They shall include material certifications, manufacturers' literature, payrolls, and work schedule.
- B. Requirements: Refer to specifications for submittals required. Allow at least 5 working days for review. Submittals shall be delivered to the Contracting Officer for review.
- C. Identification of Submittal: Submit two copies of each submittal for approval. Completely identify each submittal by showing at least the following information:
 - 1. Name and address of submitter, plus name and telephone number of the individual to contact for further information.
 - 2. Name of project as it appears in these specifications.
 - 3. Contract number, drawing number, and specification section number to which the submittal applies.
 - 4. Whether this is an original submittal or re-submittal.
 - 5. Each item shall clearly note the manufacturer's name and address, trade name, product, lot, style, color, catalog designation or model number, and locations of use.

1.05 CERTIFICATES OF CONFORMANCE:

- A. Requirements: The Contracting Officer may permit use, prior to sampling and testing, of materials when accompanied by Certificates of Conformance. Materials used based on a Certificate of Conformance may be sampled and tested. Installation of materials based on Certificates of Conformance shall not relieve the Contractor of responsibility for incorporating materials which conform to the requirements of the drawings and specifications. Material not conforming to those requirements will be subject to rejection, whether in place or not.
- B. Format: The form of the Certificate of Conformance provided by the supplier shall be as follows:
 - 1. The certificate shall state that the named product conforms to the Contract requirements.
 - 2. The certificate shall either be accompanied with a certified copy of the test results or certify that such tests are on file with the manufacturer and will be furnished to the Contracting Officer on request.
 - 3. The certificate shall provide the name and address of the manufacturer, the testing agency, and the date of tests.
 - 4. The certificate shall set forth the means of identification which will permit field determination of the product delivered as being the product covered by the certification.

1.06 QUALITY OF ASSURANCE:

- A. Codes and Standards: The work shall comply with codes and standards applicable to each type of work and as listed in the individual sections of these specifications. This Contract incorporates materials, applications, and tests by reference, with the same force and effect as when they were given in full text.
- B. Conflict: Where a conflict occurs between reference documents and project specifications, the project specifications shall govern.

1.07 DELIVERY, STORAGE, AND HANDLING:

- A. Delivery: The Contractor shall protect products, incorporated into the work, from damage while in transit to the site. Products must be delivered in original unopened containers with manufacturer's name and brand designation and contents legibly indicated. The containers shall bear the referenced specification number, type, and class as applicable.
- B. Storage: The Contractor shall provide temporary storage facilities for products. Storage shall comply with manufacturer's instructions which will protect them from damage until installation on the project and final acceptance. The storage area shall permit access for inspection and handling.
- C. Handling: The Contractor shall load and unload products in such a manner that protects them from damage until installation on the project.

1.08 PROJECT/SITE CONDITIONS:

- A. Access to the Work:
 - 1. Access to the work shall be provided by and at the Contractor's expense. No building of access roads will be allowed.
 - 2. Public or private access roads damaged by the Contractor shall be restored, at the Contractor's expense, to the same condition they were in at the commencement of work.
 - 3. Access to construction shall be accomplished with regard for environmental considerations. Construction activity may be restricted during wet or frozen conditions. Mechanized equipment use will be avoided if equipment causes rutting to a depth of 4 inches or more.
- B. Environmental Considerations:
 - 1. Operations of the Contractor (including storage of materials, project office, camping area, and equipment parking) upon Government premises shall be confined to areas approved by the Contracting Officer. The surrounding landscape shall not be disturbed. If it is disturbed, the Contractor shall restore that area at the Contractor's expense.
 - 2. Cattle in Area: The Range Management Specialist and the Permittee will coordinate to ensure cattle in the area will not affect fence construction.

3. Vehicles and equipment shall be washed and cleaned of plant seeds prior to entering the project area to prevent the introduction of invasive plants/seeds.
4. Work on this project can only occur between July 15th and November 30th to accommodate wildlife concerns in the vicinity.
5. Utilities: The contractor is solely responsible for contacting the Utilities Underground Location Center at Montana811.org (1-800-424-5555) and getting the construction area cleared before beginning construction. All utilities shall be marked out 100 feet on either side of any work associated with this contract. The contractor shall comply with all OSHA, or utility company protocols, whichever is more severe, when working in and around any buried or overhead utilities. The contractor will be responsible for the cost of all repairs and personal damages caused by contact with any utilities during the execution of this contract.

1.09 SEQUENCING AND SCHEDULING:

- A. Work Schedule: The schedule shall be submitted at the pre-work conference. When requested, the Contractor shall submit an updated schedule within three calendar days.
- B. Permitted Work Period: Work on this project can only occur between July 15th and November 30th during any calendar year to accommodate wildlife concerns in the vicinity.

PART 2: PRODUCTS

(There are no applicable requirements.)

PART 3: EXECUTION

3.01 PROJECT MEETINGS:

A. Pre-Work Conference:

1. Notification: A pre-work conference will be held prior to the start of work. The Contractor will be notified in advance of meeting time, date, and place. The purpose will be to review required work, project drawings and specifications, construction schedules, payroll and payments, and administrative provisions of the Contract.
2. Attendance: The Contractor, subcontractors, and the persons responsible for coordination of the work shall be present at the meeting.
2. Presentation: The Contractor shall be prepared to summarize and explain procedures planned for the project and present the submittals requested in the specifications.

B. Mobilization:

1. Preparatory work and operations, including those necessary to the movement of personnel, equipment, supplies, and incidentals to the project site; and for other work and operations to be performed or costs incurred prior to beginning work on the various items on the project site. This specification does not include moves between project sites.

C. Progress Meetings:

1. Location: Meetings shall be held at the project site, or as determined by the Contracting Officer.
2. Notification: Meetings may be called by either the Contracting Officer or the Contractor. Request shall state who should attend and include an agenda.

D. Final Inspection:

1. Location: A final inspection meeting shall be held at the project site, or as determined by the Contracting Officer.
2. Notification: The Contractor shall notify the Contracting Officer in writing at least three working days before the completion date so the Government can schedule final inspection.
2. Attendance: The Contractor shall require the superintendent to be present during this inspection.

3.02 PREPARATION:

- A. Work Layout: For new construction, the Government will set construction stakes to give the Contractor labels and approximate locations of fence assemblies necessary for the work executed under the Contract. For fence projects that are tear out and replace, it will be the Contractor's responsibility to maintain the fence alignment and assembly locations, unless shown otherwise on the drawings. Baseline monuments or benchmarks disturbed or removed during construction shall be reset by the Contractor at the Contractor's expense.
- B. Installation Changes: Records shall be maintained by the Contractor and the Government for actual installations where at variance with the work shown, where changed, or where not otherwise located by dimension, elevation, or other reference on the drawings.

3.03 PROTECTION OF EXISTING PROPERTIES:

- A. Take precautions to protect all adjoining private and public property and facilities, including underground and overhead utilities, structures, and fences. Restore or replace all disturbed or damaged facilities to original condition at the Contractor's expense, taking necessary steps and precautions to protect buried utilities.

3.04 DISPOSAL OF DEBRIS:

- A. Requirements: All waste, including scraps and packaging materials (empty wire spools, wooded pallets, metal bands, plastic sacks, cardboard boxes, etc.), resulting from construction of the fence shall become the property of the Contractor and shall be removed from the site by the Contractor. Disposal of materials in a legal manner and payment of fees required shall be the Contractor's responsibility. Burning of removed materials is not permitted on the work site.

PART 4: MEASUREMENT AND PAYMENT

4.01 METHOD OF MEASUREMENT:

- A. Units: The work described in this section will not be measured for payment.

4.02 BASIS OF PAYMENT:

- A. Payment: No direct payment for the work described under this section will be made. No modification and adjustments shall be allowed for costs and changes to mobilization despite changes in fuel or transportation costs. No direct payment for moves between project sites will be made. The Contractor shall include consideration for items in this Section in the bid price for other items of the Contract.

END OF SECTION 01009

PART 1: GENERAL

1.01 SUMMARY:

- A. Section Includes: Aligning and installing with Contractor-furnished property, wire fences, various panels, sage grouse fence markers, and gates as shown on the drawings and on the Work Data Sheet, and as specified herein at locations shown on the maps unless staked otherwise.

1.02 REFERENCES:

- A. American Society for Testing and Materials:

ASTM A121-22	Zinc-Coated (Galvanized) Steel Barbed Wire
ASTM A641-19	Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM A702-13(18)	Steel Fence, Posts, and Assemblies, Hot-Wrought.
ASTM A53-22	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

- B. American Wood Preservers' Association:

AWPA U1-22	User Specifications for Treated Wood
AWPA T1-22	Processing and Treatment Standards
AWPA M4-21	Standard for the Care of Preservative-Treated Wood Products
AWPA P5-15	Standards for Waterborne Preservatives

- C. If more current versions of the reference are available or become available before the date of the bid, they shall be used in place of the ones noted above.

1.03 SUBMITTALS:

- A. General: Submittals shall be according to section titled – General Information and Requirements.
- B. Certificates of Conformance: Submit one copy of written certification from the supplier of the following contractor-furnished items to be used on this project. Submit written certification from the treating plant stating that chemicals and methods used for wood preservation and net amount of preservative retained are in conformance with applicable standards. Submit written certification from the manufacturer that the Contractor-furnished fence posts and wire conform to the requirements of this section.

PART 2: PRODUCTS

2.01 CONTRACTOR-FURNISHED MATERIALS:

- A. Wire:

1. Heavy Duty Barbed Wire: (Redbrand or approved equal) Two 12-1/2 gauge galvanized wires twisted together with 2-point barbs of 14-gauge wire spaced 4-6 inches apart. Wire and barbs shall be zinc-coated steel, with a zinc coating of at least 0.28 oz/ft² of coated surface area (Class 1). The minimum breaking strength shall be 950 lb.-force. The barbed wire shall be a heavy-duty wire

conforming to ASTM A121 requirements. Failure to meet any of these requirements shall result in the rejection of the material at the time of inspection.

2. Heavy Duty Barbless Wire: Two 12-1/2 gauge galvanized wires twisted together. Wire shall be zinc-coated steel wire of at least 0.28 oz/ft² of coated surface area (Class 1) and a minimum breaking strength of 950 lb.-force. The wire shall be heavy-duty with a high tensile strength, low stretch, tight braiding per rod, conforming to ASTM A641 requirements and must be tagged accordingly. Failure to meet any of these requirements shall result in the rejection of the material at the time of inspection.
3. Single Strand Barbless Wire: Shall be 9-gauge steel with a zinc-coating of at least 0.28 oz/ft² of coated surface area (Class 1) and a minimum breaking strength of 950 lb.-force. This wire shall conform with ASTM A121 requirements.
4. Brace Wires: Shall be 4 strands of 9-gauge barbless galvanized wires, or two 12-1/2 gauge double twisted galvanized barless wires twisted tight (total of 4 strands). This wire shall conform to with ASTM A121 requirements. Brace wires should be pre-stretched before using.
5. Miscellaneous Wire: Wire for gate loops shall be 9-gauge smooth or 12-1/2-gauge Class 1 galvanized barbless wire.

B. Posts and Braces:

1. Steel Brace Posts: Shall be a 2-7/8 inch outside diameter (OD) by 0.217-inch minimum wall thickness, Oil Country Tubular Goods (OCTG) steel pipe. Pipe shall be used/recycled structural OCTG casing or drill pipe in sound structural condition, clean and radiation free (<12 µR/hr.), or new reject pipe, or unused surplus pipe. Surface rust is acceptable; however, any material that is excessively pitted, rusty, flaking, peeling, or corroded will be rejected. Pipe with splits, bends, or thin spots (hot spots) will not be accepted. Pipe shall be free from oil debris. Concrete-lined pipes, pipes lined with synthetic materials, or pipes with holes or perforations will not be accepted. Remove all drill stem threads and collars. Round post caps shall be securely field installed on all steel brace posts.
2. Steel Diagonal Brace Rails: Shall be a minimum overall length of 8 feet and no greater than 9 feet 6 inches, singular piece of round tubular steel constructed with a minimum of 2-3/8-inch outside diameter (OD) by minimum 3/16-inch wall thickness. Brace rail ends shall be fabricated by one of the following two methods: 1) pinch end of brace rail and weld on 1/4-inch minimum thick steel plates; or 2) press brace rail flat on both ends. Drill or punch a hole at each end of brace rail for 3/8-inch bolt. Equip ends with capability of being mounted diagonally on all bracing structures. Brace banding shall be 2-7/8-inch heavy duty brace band, measured 1-inch wide by minimum 1/8-inch wall thickness and attached with 3/8-inch galvanized grade 5 carriage bolts, 1-1/2-inch to 2-inch in length with grade 5 galvanized flange nuts. The pre-manufactured 1-7/8 inch diameter OD x 8 foot long 11-gauge Super Brace made by Montana Fence is an acceptable product for this application.
3. Heavy Duty Steel Line Posts (T-Posts): Shall be heavy-duty grade steel posts painted green with white, silver, or lime green tops, with a welded or riveted anchor plate. Furnish with clip-type wire fasteners (punched tabs for fastening wires are not acceptable). Steel posts shall be manufactured from wrought, rail, or new billet steel, and shall have a minimum weight of 1.33 lb./lineal ft. exclusive of the anchor plate, which shall weigh a minimum of 0.67 lb. ±5% and shall be a minimum of 18 square inches in area. Length shall be as shown the Work Data Sheet. Steel fence posts and fasteners shall be in accordance with ASTM A702 requirements.

4. Wood Single Set Posts and Gate Stays: Wood posts and gate stays shall be sound single-stem members of round cross section. Single Set wood posts shall be 6 inches minimum diameter at the small end and lengths are as shown on the Work Data Sheet. Gate Stays shall be 3 inches minimum diameter at the small end and length shall be minimum 4 feet. The following species are acceptable if treated for full length: Ponderosa pine, Lodgepole pine, and Douglas-fir. Honey Locust, western red cedar (full stem with bark removed), Juniper, Osage Orange, and White Oak are acceptable without treatment. Posts are not acceptable if a straight line drawn from the geometric center of the top to the geometric center of the butt of the post falls outside of the body of the post, or at a point 2 inches or more from the center of the post. Posts that are twisted, rotted, excessively bent, have more than one check, or have a check more than half their thickness are considered unsound and are not acceptable. Posts and stays that are charred are not acceptable. See Work Data Sheet for the required single set post lengths and the depths to be set.

C. Gates:

1. Wire Gates: Shall have the same type of wire and wire spacing as the fence as shown in the drawings unless otherwise stated on the Work Data Sheet. Use wood stays for intermediate gate stays. Panels installed on each side of the gates shall be paid as a separate installation item.

D. Drill Stem Mechanical Gate Closers: Assembly shall include:

1. Gate Closer: A36 rolled steel bar 3/16-inch to 5/16-inch diameter, or adjustable minimum 3/16-inch diameter galvanized steel cable wire loop with a minimum Working Load Limit (WLL) of 840 lb.-force and Minimum Breaking Load (MBL) of 4200 lb.-force, fastened to a cam lever device. The pipe cover shall be constructed of a minimum 11-gauge steel plating made to rest upon and exceed the diameter of the 2-7/8-inch steel brace post. The mechanical break over latch shall be attached to a minimum of 2-3/8-inch OD by 11-gauge minimum wall thickness tubular steel stem that slides securely into the 2-7/8-inch OD steel brace post a minimum of 6-inches. The closer must be capable of fitting over a minimum 3-inch diameter wood gate stay and securely holding the gate stay in place. Alternatives require prior approval from the Contracting Officer with submission of samples.
2. Gate Loop: A minimum 3/16-inch diameter galvanized steel cable wire loop with a minimum WLL of 840 lb.-force and a MBL of 4200 lb.-force, securely connected to the steel brace post with a 3-inch U-bolt muffler clamp assembly 6-inches from the ground and shaped so that the minimum 3-inch diameter gate stay will be securely held in place. The muffler clamp shall be secured to the steel brace post with 9/16-inch grade 5 galvanized round U-bolt and galvanized flange nuts. A sleeve shall be welded to the muffler clamp, parallel to the leg of the U-bolt. Another sleeve shall be welded parallel to the other leg of the U-bolt, on the same face of the muffler clamp. The sleeve shall be large enough to keep terminated ends from passing through when the loop is under pressure. The galvanized steel cable shall be threaded through the sleeves to form a loop without a bind, and the ends shall be crimped with a flattened 9/16-inch grade 5 galvanized nut further secured by a weld. Alternatives require prior approval by the Contracting Officer with submission of samples.

- E. Round Post Caps for Steel Brace Posts: Galvanized steel chain-link fence dome cap (Fits common post measurements, 2-7/8-inch; Actual 3-inch by 1.57-inch low gloss) or equivalent. Fasten as recommended by manufacturer to ensure a permanent connection.

- F. Plastic Round Post Plug: Shall be made of heavy-duty thermoplastic elastomers (TPE) tapered rib style with at least 3 horizontal ribs. The top flange to measure 2-7/8-inch to 3-inch diameter. The outside diameter of the horizontal ribs must be a minimum of 2.34-inch. Holder length (measured from under top flange to bottom cap) must be at least 1.2-inch long.

G. Fasteners:

1. Staples: Shall be at least 9-gauge, barbed, bright-finish or galvanized, have slash points, and be 1-3/4 inches in length.
2. Screws: Shall be #14 heavy-duty, heat-treated carbon steel self-tapping structural construction screws with exterior-grade corrosion resistant coating (6-inches long, 0.25-inch shank diameter).
3. Clips: Shall be 11-gauge galvanized wire.

H. Post Grout: The grout shall be a rapid setting commercial product manufactured for anchoring posts.

I. Fasteners, bolts, nuts, washers, and other accessories shall be galvanized, or cadmium plated.

J. Stakes: Wooden stakes shall be used to identify fence alignment.

K. Paint: Fluorescent orange paint shall be used to mark stakes depicting the fence/assembly alignment.

L. Surveying Equipment: Survey-quality equipment shall be used to establish alignment of the fence line and assemblies.

M. Wood Preservative: Copper Naphthenate shall be used for dressing field cuts.

N. Sage Grouse Fence Markers: Fence markers shall be made of vinyl siding undersill trim. The undersill trim shall have a lip that can be snapped onto barbed wire, with the ability to slide between barbs, but stay securely on the wire.

1. Color of vinyl undersill trim shall be white.
2. Depth of trim shall be 1-5/8" or greater.
3. Length of each fence marker shall be 3-inches.
3. Cut undersill trim strips with a 10-inch miter saw with a 200-tooth blade. For efficiency, strips may be stacked and cut up to 16 pieces at a time. Tin snips may be used for small projects.
4. The use of commercially produced fence markers that meet these specifications is permissible.

2.02 WOOD TREATMENT PRACTICES:

A. General: The following treatments may be used. Minimum net retention and penetration shall be according to AWP A U1, Section B, and shall meet Standard UC4A. Treatment method shall be according to AWP A T1, except where otherwise noted.

B. Chromated Copper Arsenic (CCA): Waterborne preservative according to Type A, B, or C in AWP A P5.

C. Pentachlorophenol Method: Five percent pentachlorophenol solution conforming to AWP A P35, Type A.

D. Creosote: Shall be according to AWP A P2.

- E. Field Cutting: Shall be treated with copper naphthenate according to AWP4 M4. When treated, member must be bored or cut during construction, thoroughly swab untreated surfaces.

PART 3: EXECUTION

3.01 PREPARATION:

- A. Clearing: Clearing shall be limited to cutting brush debris, such as big sage, greasewood, pine, or juniper type species in the actual fence line alignment. Maximum clearing width is 6 feet on each side of the fence line. Provide minimum disturbance to existing grass and sod. Clearing will be attained by mowing vegetation or with a chainsaw for brushing or tree removal. Trees removed will be lopped and scattered to a maximum height of 18 inches above ground level. Stumps will be left no taller than 2 inches above ground level. No additional clearing of the fence line shall be allowed without prior approval of the Contracting Officer. Road construction shall not be allowed. Traffic or equipment shall be confined to routes approved by the Contracting Officer. Travel along the newly constructed fenceline shall be limited to UTV's and skid steers.

3.02 INSTALLATION:

- A. Alignment: For new construction, the final fence alignment and assemblies shall be aligned and staked by the Contracting Officer. Stakes shall be set at a minimum distance of 500 feet or as approved by the Contracting Officer. Approximate locations of the assemblies will be staked by the Government, and the Contractor shall align all assemblies within each run of fence. The lateral alignment shall be within plus or minus 1/2 inch. For tear out and replace construction, Contractor shall be responsible to stake existing fence alignment and assemblies prior to tear out, unless otherwise shown on the drawings. Contractor is responsible for the cost of relocating lost original fence alignments and assemblies.
- B. General: Steel brace posts shall be used for all End Panel, Corner Panel, Stress Panel, and gate-panel posts. Panels shall be installed in accordance with the Work Data Sheet and the drawings, and as directed by the Contracting Officer. Wood posts shall be used for all single set posts and shall be installed in dug or drilled holes. Post spacing shall be within 6 inches, but only when ground conditions prevent exact spacing. When treated wood members must be bored or cut during construction, thoroughly swab untreated surfaces with approved preservative. Avoid breathing sawdust and direct contact of preservative on the skin during field cutting and swabbing. Refer to the preservative manufacturer's required warnings for handling treated members.
- C. Setting Single Set Wood Posts: This bid item is to be used to construct auxiliary fence panels in drainage crossings or at major grade changes along the fence line. Installation shall be in accordance with the Work Data Sheet and the drawings. The Contractor shall dig or drill holes for setting single set wood posts to the designated depth as shown on the Work Data Sheet. Set posts plumb and to the spacing and grades as shown on the drawings, unless staked otherwise. Lateral alignment shall be within plus or minus one-half inch. Dug or drilled holes shall provide adequate open space around the post so that backfill can be tamped the full depth around the post. Backfill gradually and uniformly with soil around each post. Do not exceed 6-inch lifts during backfill. Compact backfill firmly from the bottom of the hole to the ground surface. When rock formations are encountered, dig or drill holes a minimum of 30 inches deep and slightly larger than the diameter of the post. Place the post into the hole and grout post solidly in position with cement grout or mortar.

*Note: Milled single set wood posts shall be allowed to have an anchor plate or block attached to the bottom or a wood block attached to the side of the post base area to anchor the post in place since it has no taper to properly hold it in place.

- D. Ratio of Wood to Steel Line Posts: Line posts shall be all steel T-Posts except as noted on the drawings and within these specifications.
- E. Driving Steel Posts:
 - 1. Steel Brace Posts: Drive into the ground to the depth shown on the Work Data Sheet. Posts shall be installed plumb, to the designed depth, and the lateral alignment shall be within plus or minus one-half inch.
 - 2. Steel Line Posts (T-posts): Drive into the ground to depth shown on the Work Data Sheet, or until anchor plate is slightly below the ground surface. Posts shall be installed plum, to the designed depth, and the lateral alignment shall be within plus or minus one-half inch. When rock formations prevent driving, remove anchor plate of the T-post and excavate or drill hole a minimum of 18 inches deep and slightly larger than the diameter of the post. Place the post into the hole and grout the post solidly in position with cement grout or mortar.
- F. Cap all Steel Brace Posts for permanent waterproofing of the post. Steel caps shall be installed and fastened as per manufacturer's recommendations.
- G. Corner Panel, End Panel, and Stress Panel Assemblies: Construct as shown on the drawings or as staked in the field. Rock jacks shall not be substituted for these items. Construct stress panels a maximum of 80 rods center-to-center for barbed wire sections. In addition, stress panels are required at points between which wire is to be stretched. Set posts as specified, fabricate panels as shown on the drawings and indicated on the Work Data Sheet.
- H. Wire: Stretch tightly and securely attach to steel posts or staple to wood posts with standard wire clips. Wire is properly stretched when it is springy to the touch. Terminate wire at each end post, gate post, corner post, or stress panel. Wrap wire around the post two times and tie off by wrapping around the incoming wire a minimum of four times. Stretch tightly and securely attach to steel posts or staple to wood posts with standard manufactured wire clips. Drive staples into wood until the staple contacts the wire against the post, but not so tight as to crimp the wire or prevent free movement of the wire between the post and the staple. Do not drive staples parallel to the grain of the wood.
- I. Wire Spacing: The wire spacing will be measured from the ground up. Bottom wire shall be smooth unless designated otherwise. Spacing in indicated on the Work Data Sheet.
- J. Wire Clips: Wire clips shall be used on every wire at steel line post locations. Secure wire to steel posts by bending clip ends over or around wire, crimping ends to secure wire on both sides of the post, but not so tight as to prevent free movement of the wire through the clip. Each side of the clip shall be twisted around the wire and back parallel to the clip itself, (minimum 180 degrees).
- K. Gates: Shall be installed in accordance with the Work Data Sheet and the drawings. Wood gate stays shall be used at each end, with two wood intermediate stays. Alternate methods shall have prior approval from the Contracting Officer.
- L. Mechanical Gate Closures: Line up the latch with the gate. Adjust latch assembly after installation so that the latch does not bind, and the gate hangs properly.

- M. Drainage Crossing: Construct drainage crossings in accordance with the Work Data Sheet and the drawings. Construct an auxiliary fence of wire, Deadman, and single set wood posts independent of the main fence, so that if the crossing washes out, the main fence will not be damaged. This bid item is for installation of wire and Deadman only. End Panels, Three-post Corner Panels, and Single Set Wood Posts are separate bid items.
- N. Deadman: A Deadman is required when the distance between the bottom wire and the ground exceeds 20 inches. It shall be constructed of 12-1/2-gauge barbless wire bent in two and wound around the four wires in the fence and attached to full-length steel post. Steel post shall be driven 4 feet underground at a 45-degree angle with 1-1/2 feet extending above ground level. In a severe depression, as determined by the government official, two steel posts shall be used and shall be positioned opposite one another to form a 90-degree angle between the two posts for maximum holding power. Use two strands of 9-gauge barbless wire in a severe depression. Used posts 5'-6" in length or longer will be acceptable so long as they are free of corrosion and rust.
- O. Fence Stays: Wire fence stays are not permitted.
- P. Sage Grouse Fence Markers: Fence markers shall be installed on top wire and third wire of fence. Top wire shall have a minimum of four (4) markers, and the third wire shall have a minimum of three (3) markers. Markers shall be equally spaced between line posts.
1. Install fence markers over entire length of fence, with a minimum of seven (7) markers per rod.
 2. Fence markers shall be snapped securely into place between barbs such that they are able to swing freely and slide between barbs but stay secure on the wire.

PART 4: MEASUREMENT AND PAYMENT

4.01 METHOD OF MEASUREMENT:

- A. Units: The work described in this section will be measured and paid for by the rod for each type of fence as measured along the ground surface next to the completed fence, including distance across gate openings, end, stress, and corner panels. The auxiliary fence in the drainage crossings will be measured and paid for by the rod measured along the ground surface next to the completed auxiliary fence. The quantities of single set posts, end, stress, corner panels, and gates to be paid for will be the actual number of these items furnished, placed, and accepted. Note: Gates and drainage crossings do not include adjacent end or corner panels. These are separate items. Deadman is not a pay item, and the price of this item shall be considered incidental and is to be included in the bid price of other items in this section. Sage grouse fence markers are not a pay item, and the price of this item shall be considered incidental and is to be included in the bid price of other items in this section.

4.02 BASIS OF PAYMENT:

- A. Payment: Prices and payment will be full compensation for the work described in this section. No direct payment will be made for mobilization. The price for mobilization and the price of sage grouse fence markers shall be included under Pay Item MT02834(1) Barbed Wire Fence, 4-Wire.

Payment will be made under:

<u>Pay Item</u>	<u>Pay Unit</u>
MT02834(1) Barbed Wire Fence, 4-wire	Rod
MT02834(5) Barbed Wire End Panel	Each
MT02834(6) 3-Post Barbed Wire Stress Panel	Each
MT02834(9) 3-Post Barbed Wire Corner Panel	Each
MT02834(23a) Barbed Wire Gate, 4-wire	Each
MT02834(30) Single Set Wood Post	Each

END OF SECTION MT02834

DIVISION 2
SITE WORK

SECTION MT02836
FENCE DRAINAGE CROSSINGS

PART 1: GENERAL

1.01 SUMMARY:

- A. Section Includes: Furnishing and installing drainage crossings at the locations shown on the drawings unless staked otherwise.
- B. Related Sections:
General Information and Requirements Section 01009
Wire Fences and Gates Section MT02834

1.02 PROJECT/SITE CONDITIONS:

- A. Requirements: Access roads shall not be constructed along fence lines and to the construction site unless written approval is granted by the Contracting Officer. Cross country travel with wheeled or tracked vehicles shall be only to and along fence lines.

PART 2: PRODUCTS

2.01 MATERIALS:

- A. Wire Fence Materials: Shall be according to Section MT02834 - Wire Fences and Gates and as shown on the drawings.

PART 3: EXECUTION

3.01 PREPARATION:

- A. Clearing: No clearing of fence line shall be allowed, except as approved by the Contracting Officer. Provide minimum disturbance to existing grass and sod. Traffic and equipment shall be confined to routes approved by the Contracting Officer.

3.02 INSTALLATION:

- A. Drainage Crossings: Construct drainage crossings as shown on the drawings, with an auxiliary fence of wire, Deadman, and posts constructed under and independent of the main fence, so that when the crossing washes out, the main fence will not be damaged.
- B. Material: Drainage Crossings will be constructed with single set wood posts.
- C. Deadman: A Deadman is required when the distance between the bottom wire and the ground exceeds 20 inches. It shall be constructed of 12-1/2-gauge barbless wire bent in two and wound around the four wires in the fence and attached to full-length steel post. Steel post shall be driven 4 feet underground at a 45-degree angle with 1-1/2 feet extending above ground level. In extreme conditions, as determined by the government official, two steel posts shall be used and shall be positioned opposite one another to form a 90-degree angle between the two posts for maximum holding power. Use two strands of 9-gauge barbless wire in severe depression. A used steel post 5'-6" in length or longer may be used as a

Deadman so long as the post is competent and free from any deterioration.

PART 4: MEASUREMENT AND PAYMENT

4.01 METHOD OF MEASUREMENT:

- A. Units: The work described in this section is for the wire and Deadman only, all other features, such as end panels, single set wood posts and corner panels are counted and paid for separately.

4.02 BASIS OF PAYMENT:

- A. Payment: Prices and payment will be full compensation for the work described in this section.

Payment will be made under:

Pay Item

Pay Unit

MT02836(1) Drainage Crossing

Rods

END OF SECTION MT02836

PART 1: GENERAL

1.01 SUMMARY:

- A. Section Includes: Furnishing and installing two Commercially Manufactured cattle guard grids, wings, pre-cast concrete foundations, and accessories as shown on the drawings and as specified herein at locations shown on the maps unless staked otherwise.

1.02 SUBMITTALS:

- A. General: Submittals shall be according to Section 01009 - General Information and Requirements.
- B. Manufacturer's Literature: Submit three copies of the manufacturer's descriptive data including design data for commercial cattle guard grids with wings.
- C. Certificates: Submit three copies of written certification from the supplier of the Contractor-furnished cattle guards that the cattle guards to be used on this project conform to the requirements of this specification section.
- D. Precast concrete cattle guard foundations as manufactured according to designed specifications.
- E. Alternate Design: Submit three copies of detailed engineering drawings and material specifications and fabrication instructions. Alternate designs shall be furnished with design calculations and be certified by a licensed Professional Engineer that they meet design requirements.

PART 2: PRODUCTS

2.01 MATERIALS:

- A. Cattle guard grids: Provide cattle guard grids as per standard Bureau of Land Management design and as shown in drawings. Design loading of grids shall conform to AASHTO H-20 specifications. New materials shall be used in the fabrication of the cattle guards.
 - 1. Hot-rolled structural steel sections: ASTM A36.
 - 2. Cold-rolled rails: ASTM A242. This type of rail shall be welded with E7015 or E7016 electrodes.
 - 3. Structural tube sections: ASTM A500, Grade B.
 - 4. Steel pipe: ASTM A53, Schedule 80.
 - 5. Zinc-plated connectors, bolts, and nuts: ASTM A307, or SAE Grade 8: SAE J429.
 - 6. Anchor clips: Shall be furnished loose and field welded or bolted.
 - 7. Galvanized parts: ASTM A123 for steel shapes and ASTM A153 for fasteners.
- B. End Wings: Shall be as shown in drawings.

- C. Paint: Color shall be Federal Safety Yellow or match color no. 13655, Fed. Spec. TT-C-595. Cattle guards and wings shall be painted after fabrication.
- D. Foundations: Cattle guard foundations shall be precast concrete as shown in drawings. Foundations shall continuously support full length of grid on each side. Footing design shall be approved by Contracting Officer.
- E. Backfill Material: Shall be native material. Backfill shall be free of rocks, stones larger than 3 inches maximum diameter, sticks, or excessive organic materials, and shall be of the best quality available onsite.

2.02 FABRICATION:

- A. Welding: Shall be accomplished with AWS E7018 electrodes, except cold-formed rails.
- B. Steel Cattle Guard Grids: Steel grids shall be all-welded construction. Welds shall be secure and complete along both edges of cross pieces and at joints in the frame and shall conform to AWS D1.1.
- C. Cattle Guard Foundations: Precast concrete foundations shall be constructed as specified and shown on the drawings. Alternate designs shall be approved by the Contracting Officer.

PART 3: EXECUTION

3.01 PREPARATION:

- A. Clearing: No clearing of the fence line shall be allowed without prior approval by the Contracting Officer. Traffic or equipment shall be confined to routes approved by the Contracting Officer.
- B. The contractor is responsible for locating and protecting all underground utilities. The CALL BEFORE YOU DIG number is 1-800-424-5555. This must be done at least 3 working days before work starts. The contractor is responsible for all costs of repairing damaged utilities, if any occur.

3.02 INSTALLATION:

- A. Precast Concrete Foundation: Concrete foundations shall be centered on, and placed perpendicular to, the existing roadways.
- B. Grid: Securely bolt cattle guard grids to the bases as shown on the drawings.
- C. Wings: Securely bolt the end wings to the adjacent fence, as shown on the drawings.
- D. Gate: Install gate according to Section MT02834 – Wire Fences and Gates.
- E. Excavation and Backfill: Initial excavation for cattle guard bases may be made with power equipment, but over excavation shall be avoided. Over-excavated areas shall be backfilled and compacted, as required for all compacted fill. Provide drainage from excavated area to prevent water from pooling within the excavation, as shown on the drawings. Earthen materials removed during excavation shall be replaced along the abutment edge of the cattle guard. After installation, backfill and compact the cattle guard with native material as per this specification. Backfilled roadway material shall be compacted and the road surface finished and smoothed so as not to create a bump and shall allow for smooth travel over the cattle guard.
- C. Compaction requirements: Place backfill in layers not exceeding 6 inches in thickness. Compact each layer before placing the next layer with a mechanical tamper adequate to fit in any voids

between concrete base and unexcavated material. Each layer shall meet or exceed 95% of surrounding unexcavated material in roadway. The material shall have sufficient moisture to provide optimum compaction.

- D. Gates: Install gates according to Section MT02834 – Wire Fences and Gate.

PART 4: MEASUREMENT AND PAYMENT

4.01 METHOD OF MEASUREMENT:

- A. Units: The work described in this section will be measured and paid for as a complete cattle guard assembly including grids, wings, precast concrete footings, and other materials or supplies necessary for the installation of the cattle guard unit. All required excavation, embankment and grading work will be incidental to each separate pay item.

4.02 BASIS OF PAYMENT:

- A. Payment: Prices and payment will be full compensation for the work described in this section.

Payment will be made under:

<u>Pay Item</u>	<u>Pay Unit</u>
MT02879(13) Cattle Guard with Wings and Footings	Each

END OF SECTION MT02879

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WORK DATA SHEET
CK CREEK PASTURE FENCE

SECTION MT02834 - WIRE FENCES AND GATES

BARBED WIRE FENCE, 4 WIRES

4 Wire Fence - Steel

Line Posts Wire Spacing:

E: 8 inches – Top of post
D: 12 inches – Barbed wire
C: 6 inches – Barbed wire
B: 6 inches – Barbed wire
A: 16 inches – Smooth wire

4 Wire Fence – Single Set Wood Posts

Single Set Posts Wire Spacing:

E: 8 inches – Top of post
D: 12 inches – Barbed wire
C: 6 inches – Barbed wire
B: 6 inches – Barbed wire
A: 16 inches – Smooth wire

Measure spacings from ground up. See drawings for spacing locations.

All steel and wood post spacing: (L): 16'-6"

Length of single set wood posts (H1): 7'-0"

Depth of single set wood posts in ground (h1): 3'-0"

Diameter of single set wood posts: 6"

Length of steel line posts (T-Posts) (H2): 5'-6"

Depth of steel line posts in ground (h2): 1'-6"

Length of steel brace posts (long) (H3): 10'-0"

Depth of steel brace posts (long) in ground (h3): 5'-6"

Length of steel brace posts (short) (H4): 6'-0"

Depth of steel brace posts (short) in ground (h4): 5'-0"

Gate post spacing (standard): 16'-6"

4-Wire type fence – All wires barbed, except bottom wire is smooth, and no wire stays.

All gates shall be 4-Wire type – All wires barbed, except bottom wire is smooth.

Round post caps shall be field installed and securely fastened on all steel brace posts.

Sage Grouse fence markers shall be field installed and securely fastened along entire length of fence. Top wire shall have a minimum of 4 markers per rod evenly spaced and third wire down shall have a minimum of 3 markers per rod evenly spaced. See Line Panels Detail on Sheet 5.

DRAWING SHEET NO. 5 of 8, SPECIFICATON NO. MT02834(1) Typical FENCE (LINE PANELS)

Type of barbed wire fence – 4-wire (3 barbed, 1 smooth at bottom; no stays)

DRAWING SHEET NO. 7 of 8, SPECIFICATON NO. MT02834(5) Typical END PANEL

Type of steel post barbed wire end panel – 2-post steel end panel (1 long post, 1 short post)

DRAWING SHEET NO. 5 of 8, SPECIFICATON NO. MT02834(6) Typical 3-POST STRESS PANEL

Type of steel post barbed wire stress panel – 3-post steel stress panel (1 long post, 2 short posts)

DRAWING SHEET NO. 5 of 8, SPECIFICATION NO. MT02834(9) – Typical 3-POST CORNER PANEL

Type of steel post barbed wire corner panel – 3-post steel corner panel (1 long post, 2 short posts)

DRAWING SHEET NO. 7 of 8, SPECIFICATION NO. MT02834(23a) Typical 4-WIRE GATE

Width of barbed wire gates - 16.5' minimum

DRAWING SHEET NO. 5 of 8, SPECIFICATION NO. MT02834(30) Typical SINGLE SET WOOD POST

DRAWING SHEET NO. 6 of 8, SPECIFICATION NO. MT02836(1) – Typical DRAINAGE CROSSING

End panels and wood set posts are separate bid items.

DRAWING SHEET NO. 8 of 8, SPECIFICATION NO. MT02879(13) – Typical CATTLE GUARD with WINGS and FOOTINGS

CONTRACT ESTIMATED QUANTITIES

ESTIMATED QUANTITIES

Bid Item	CK CREEK PASTURE FENCE	Totals
MT02834(1)	Barbed Wire Fence, 4- Wire	900 rods
MT02834(5)	Barbed Wire End Panel	18 each
MT02834(6)	3-Post Barbed Wire Stress Panel	4 each
MT02834(9)	3-Post Barbed Wire Corner Panel	12 each
MT02834(23a)	Barbed Wire Gate, 4-Wire	5 each
MT02834(30)	Single Set Wood Post	100 each
MT02836(1)	Drainage Crossing	74 rods
MT02879(13)	Cattle Guard with Wings and Footings	2 each

END OF SECTION – WORK DATA SHEET

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